

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November – December 2011

EC 201 Digital Electronics & Logic Design

Date: 29.11.2011, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Represent the decimal number 8620 in BCD and 8, 4, -2, -1. [02]
- (b) The numbers that immediately follows $(677)_8$ and $(FFFF)_{16}$ are _____ and _____ respectively [02]
- (c) Why NAND and NOR gates are known as universal gates? [03]
- Q - 2 (a) Expand $A+BC'+ABD'+ABCD$ to minterms and maxterms. [04]
- (b) Design Full Subtractor with two half subtractor and OR gate. [04]
- (c) Design a combinational circuit that converts a decimal digit from 8, 4, -2, -1 code to BCD. [06]

OR

- Q - 2 (a) Using Boolean Algebra prove the following. [04]
- (i) $(B+BC)(B+BC')(B+D) = B$
- (ii) $xyz+x'y+xyz' = y$
- (b) Show how a Full Adder can be converted to a full subtractor with the addition of one inverter circuit. [04]
- (c) Design a combinational circuit that converts a decimal digit from 2,4,2,1 to 8,4,-2,-1 code. [06]
- Q - 3 (a) Simplify the function $F(A,B,C,D,E,F)=\sum(6,9,13,18,19,25,27,29,41,45,57,61)$ with either K-map or Tabulation method. [08]
- (b) Construct a 5×32 Decoder with 3×8 Decoders and 2×4 Decoder. [06]

OR

- Q - 3 (a) The Boolean expression [07]

$$BE + B'DE'$$

is a simplified version of the expression

$$A'BE + BCDE + BC'D'E + A'B'DE' + B'C'DE'$$

Are there any Don't care conditions? If so, what are they?

(b) Implement function

[07]

$$F(A,B,C,D) = \sum(2,4,6,7,9,13,15)$$

with 8×1 Multiplexer.

SECTION – II

Q - 4 (a) What is the difference between Combinational circuit and Sequential circuit? [02]

(b) Explain Master-Slave Flip flop. [05]

Q - 5 (a) Design a counter with the following binary sequence 0,1,3,2,6,4,5,7 and repeat. Use R-S flip flops. [07]

(b) Design 4 bit bidirectional shift register with parallel load. [07]

OR

Q - 5 (a) Design 4 bit Ripple Binary Counter. [07]

(b) Design a counter with the following binary sequence 0,1,3,7,6,4 and repeat. Use T flip flops. [07]

Q - 6 (a) Design 4 bit Synchronous Binary Counter. [07]

(b) Explain Johnson Counter. [07]

OR

Q - 6 (a) Design 4 Bit Ripple BCD counter. [08]

(b) Explain [06]

(1) Shift register

(2) Triggering of flipflops
